Nathan R. Carpenter

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SKILLS:

Software: HTML, CSS (Bootstrap), ReactJS, Python, MySQL, Flask, Freesurfer, Linux, Ubuntu, Meshlab, Meshmixer, Simplify3D, and Solidworks

PROJECTS:

3D Brain Modeling project May 2021 -

- Utilized Freesurfer's software to segment NIFTI formatted MRI scans
- Enhanced 3D brain models by utilizing Meshlab and Meshmixer
- Printed models using Phrozen Sonic 4K mini, Flashforge Dreamer, Flashforge Finder, Creality 300, Stratasys Mojo
- Created a full stack website for the project using HTML, CSS, Javascript, MySQL, and Flask

Survey May 2022 -

- Created a survey to collect data from the user and store it using SQL
- Flask was installed and utilized as a server that connects the webpages
- Python was utilized to redirect the user to a page with their survey input

WORK EXPERIENCE:

Summer Research Program: Stevens Institute of Technology
Soft Matter Bio-mechanics Lab Researcher May 2021-August 2021

- Spearheaded a research group by creating a roadmap of our goals
- Researched multiple segmentation software programs to learn how to create 3D brain models from MRI images
- Successfully 3D printed models of the brain

ANC Sports Enterprise, World Trade Center, New York
Field Technician September 2019-May2021

- Aided in solving any issues that were encountered with the digital display's software or hardware, by replacing scan cards and rewiring signs
- Ensured LED signs were displaying the proper content and the screens were adequately functioning to maximize our client's advertising campaigns

Union Paving Construction Company, Mountainside, NJ
Project Management Intern June 2019–August 2019

- Performed Takeoffs for a variety of different jobs to determine materials needed for potential projects and to establish a strategy to apply
- Assisted managing construction sites to ensure procedures went according to plan
- Calculated the proper excavation and fill for railroad track to maximize efficiency

ORGANIZATIONS/COMMUNITY ACTIVITIES:

- Stevens Institute of Technology Basketball Team 2017-2021
- Mechanical Engineering Student Advisory Council Member 2019-2022
- Duckling program 2017- 2018
- Hoboken Boys and Girls Club 2018-2019

EDUCATION:

Stevens Institute of Technology, Hoboken, NJ
Bachelor of Science in Mechanical Engineering, Graduation: May 2022
GPA: 2.9 Coursework: Additive Manufacturing, Computer Science, Fluid Mechanics, Multivariable Calculus, Electricity and Magnetism, Differential Equations

Coding Dojo:
Full Stack Web Development Certification, Graduation: June 2022